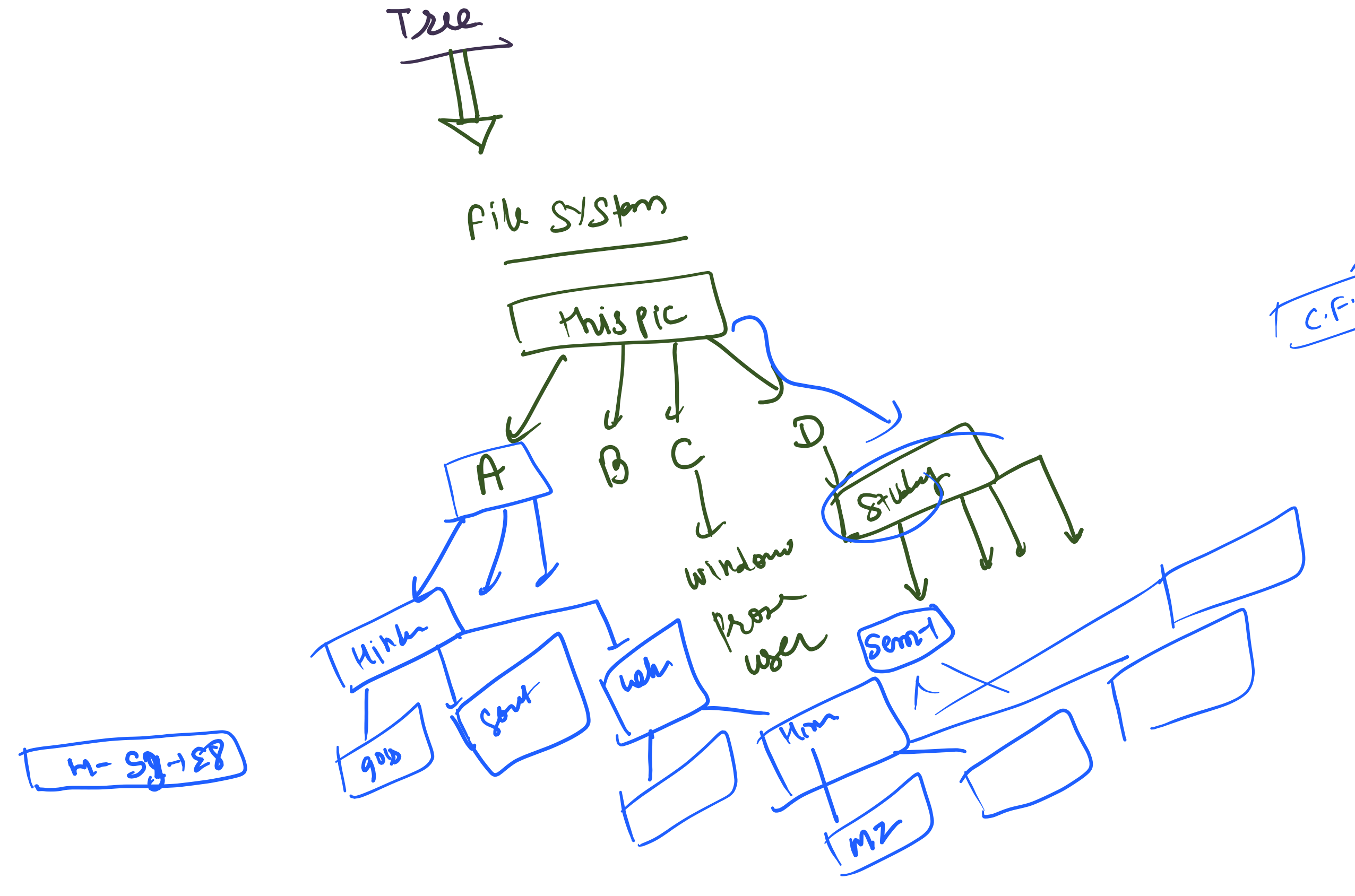
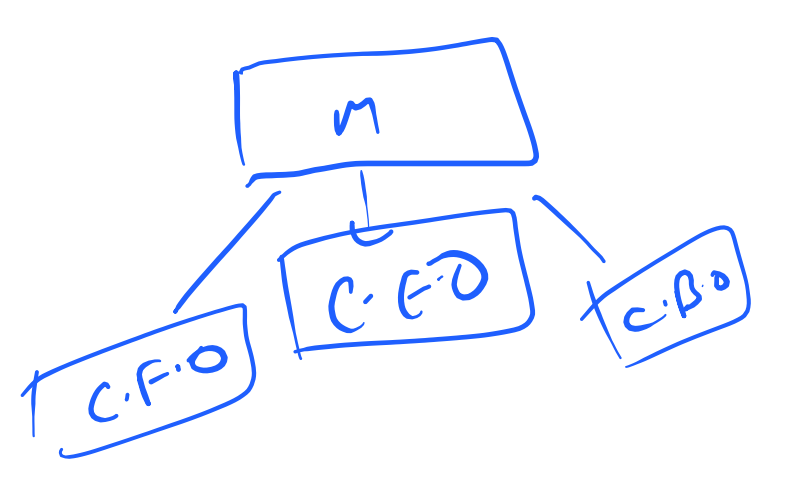
Lec-30

27/09/26 1:11 PM

Tree

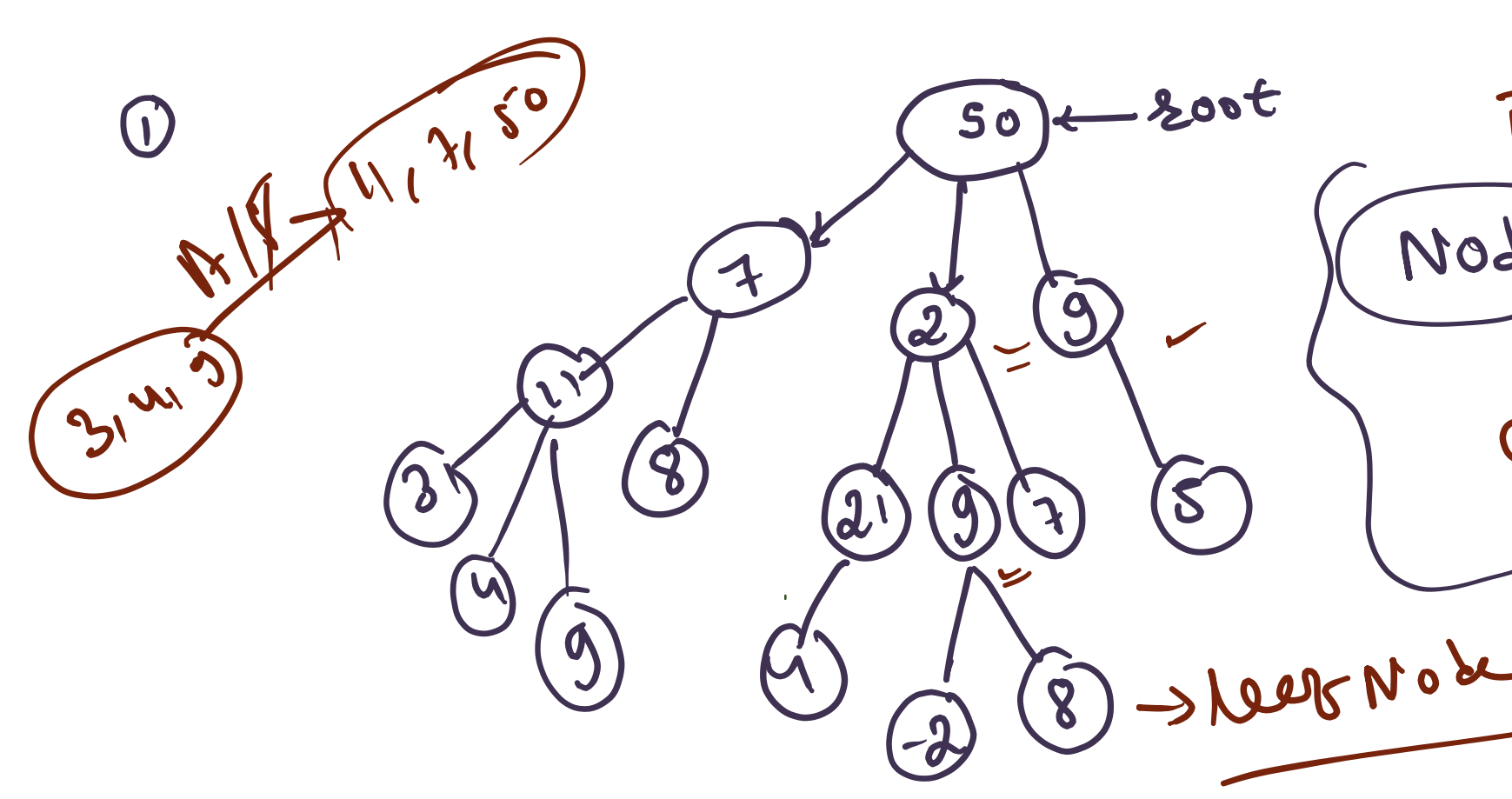

File System
this pic
A
B
C
D
Hindi
Sount
gold
web
Window
Prog
user
Study
Sem-1
Hmm
MZ
M-SG-128


M
C.F.O
C.E.O
C.B.O

(i) Generic Tree ✓
(ii) Binary Tree
(iii) Binary Search Tree
(iv) Prefix Tree
(v) Trie

99.5

①


A/P
3, 4, 9
11, 7, 50
50
root
7
2
9
11
8
21
9
7
5
3
4
9
4
-2
8
→ leaf Node

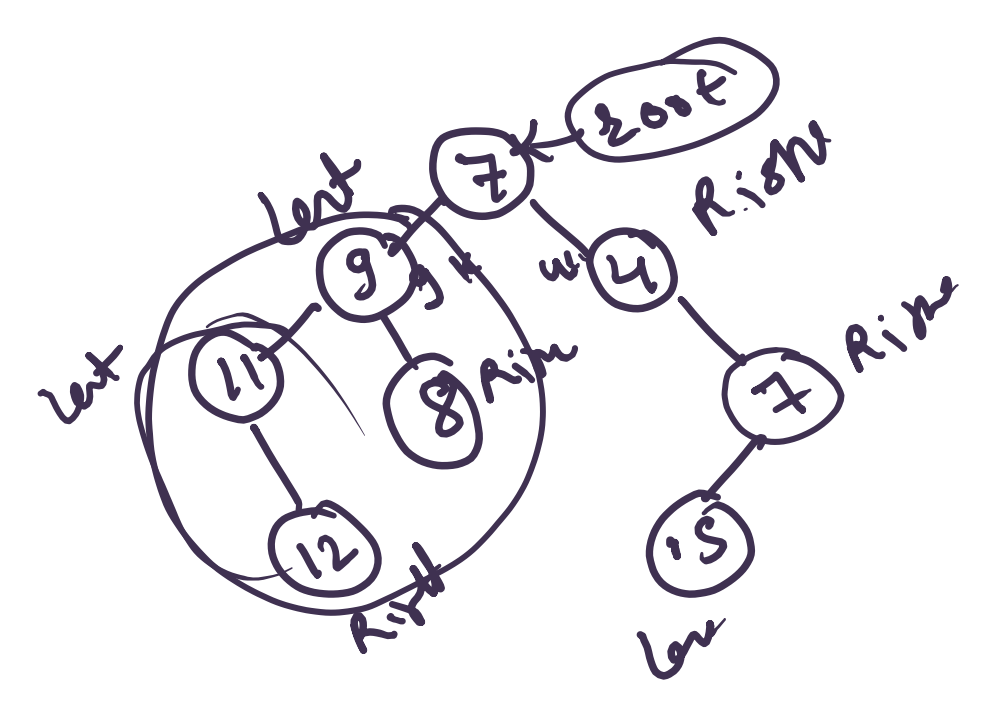
No child X

Node [इसमें atmax दो !]

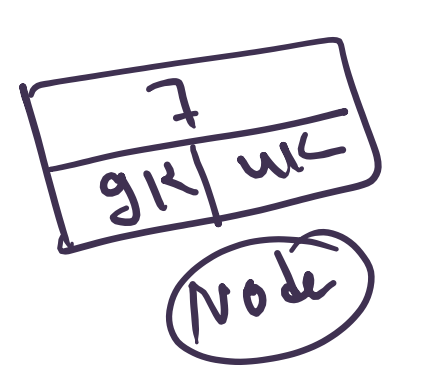
Binary Tree BT
Binary Search Tree

दो बच्चा मिट्ठी खीर
उससे ज्यादा बवासीर

class Node{
int val
Node left
Node Right

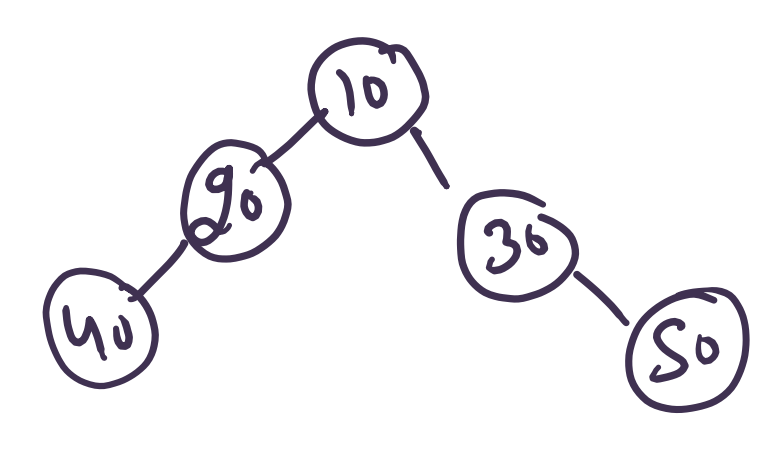

Root
7
Left
9
4
Right
Left
11
8
Right
7
Right
12
Right
15
Left

Bihar
{C.M}



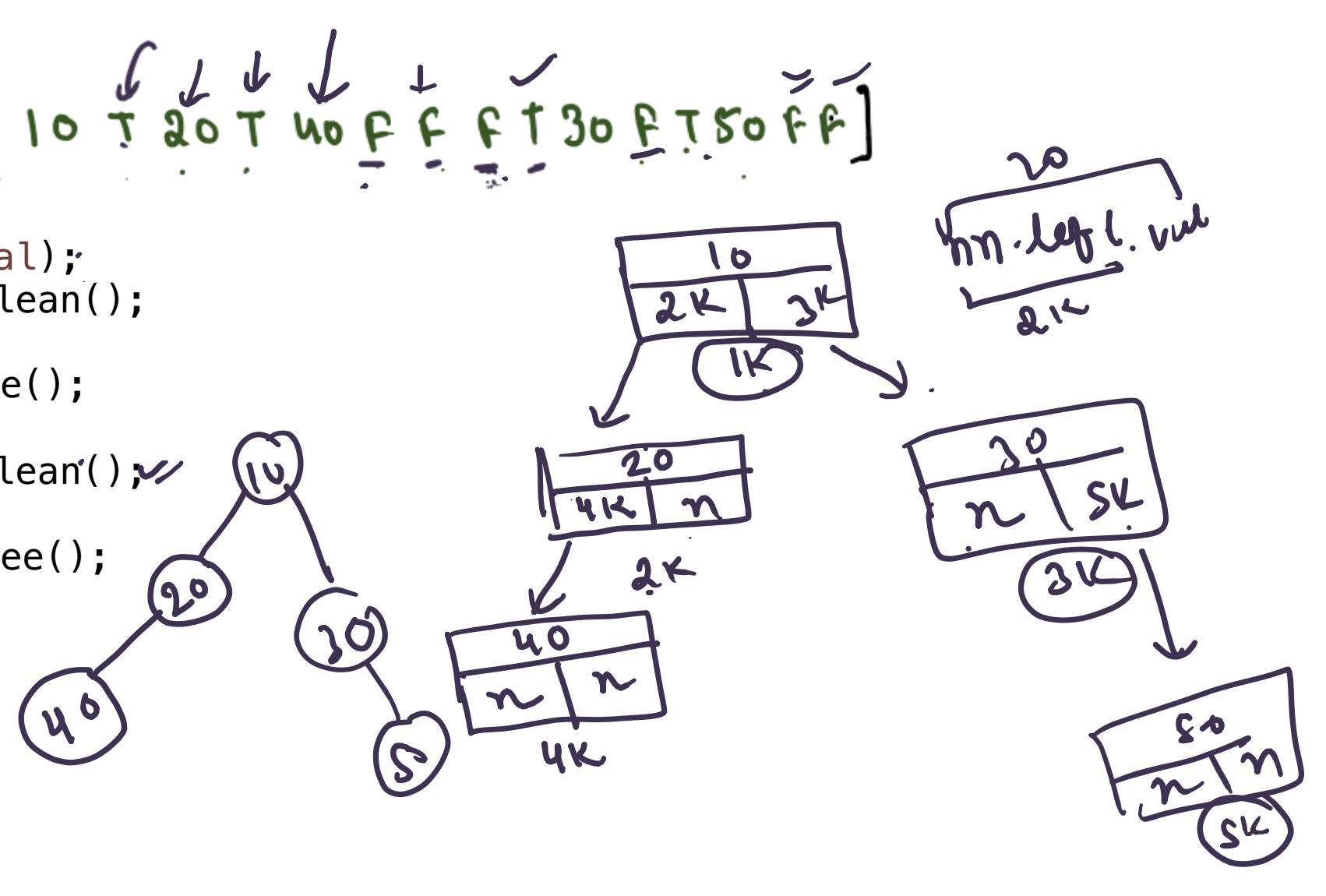
Root → Left Right

[10 T 20 T 40 F F F T 30 F T 50 F F]


10
20
40
30
50

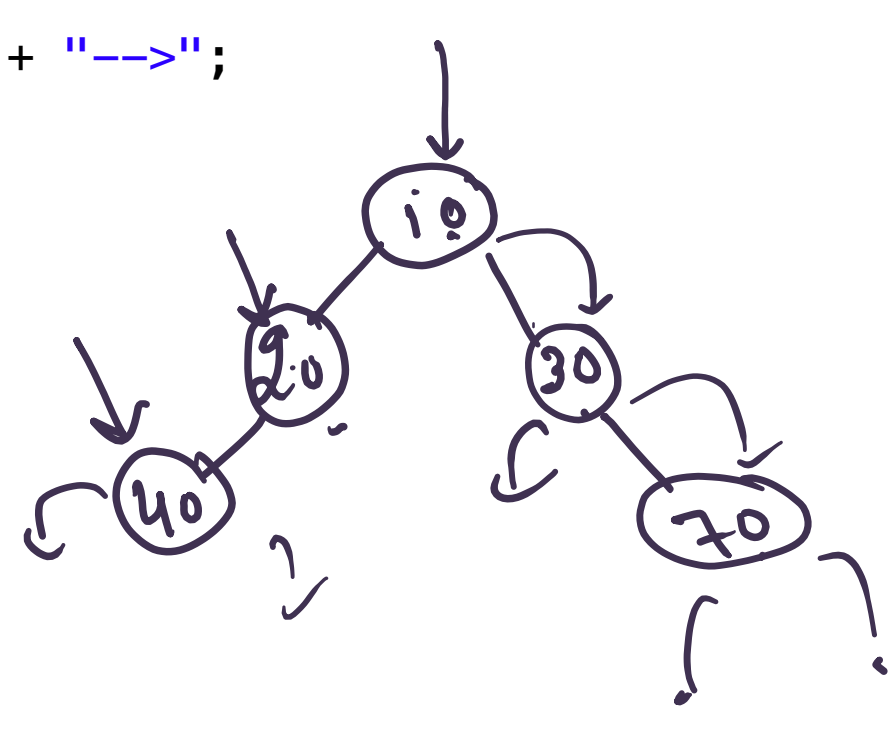
```
private Node createTree() {
    // TODO Auto-generate
    int val=sc.nextInt();
    Node node = new Node(val);
    boolean hlc=sc.nextBoolean();
    if(hlc) {
        node.left=createTree();
    }
    boolean hrc=sc.nextBoolean();
    if(hrc) {
        node.right=createTree();
    }
    return node;
}
```

[10 T 20 T 40 F F F T 30 F T 50 F F]

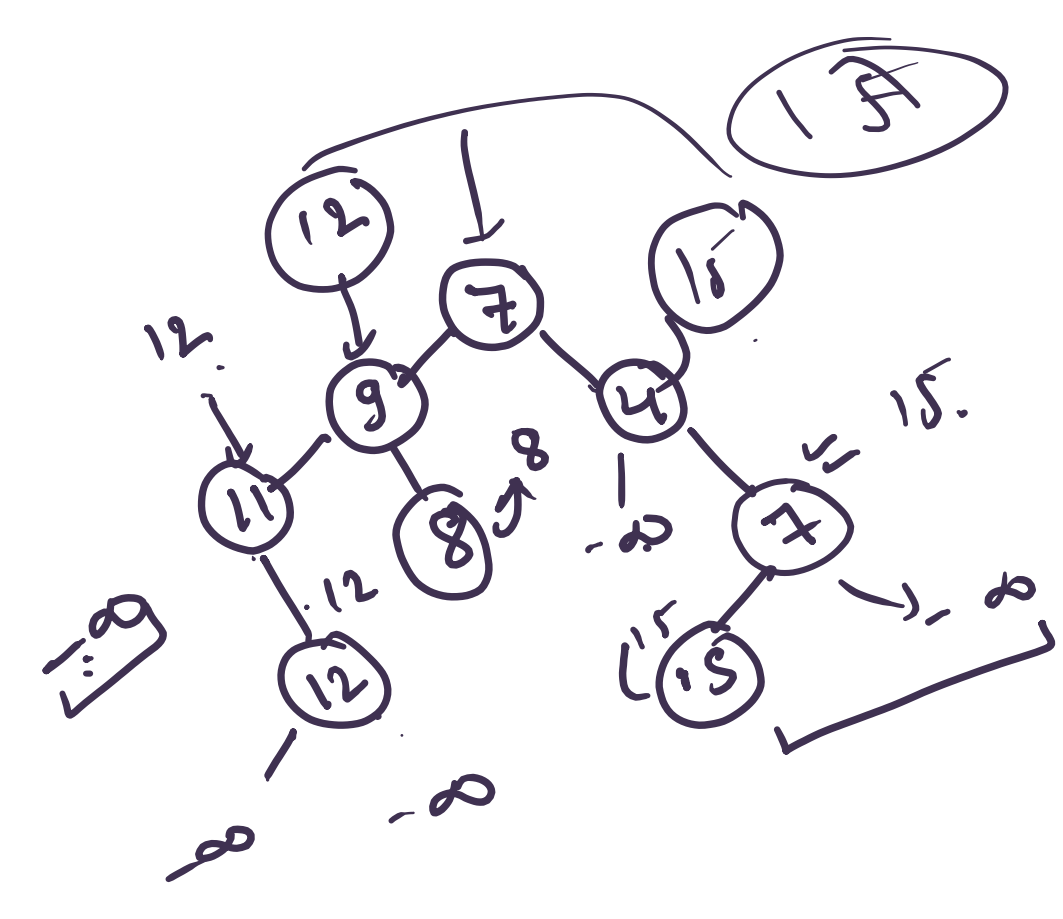

10
20
40
30
50
10
2K
3K
1K
20
4K
n
2K
40
n
n
4K
30
n
5K
3K
50
n
n
5K
20
nn.left.val
2K

20 ←10→ 30
40 ←20→ .
. ←40→ .
. ←30→ 70
. ←70→ .

```
public void Display(Node nn) {
    String s  = "<--" + nn.val + "-->";
    if(nn.left!=null) {
        s=nn.left.val+s;
    }
    else {
        s="."+s;
    }
    if(nn.right!=null) {
        s=s+nn.right.val;
    }
    else {
        s=s+".";
    }
    System.out.println(s);
    Display(nn.left);
    Display(nn.right);
}
```


10
20
30
40
70

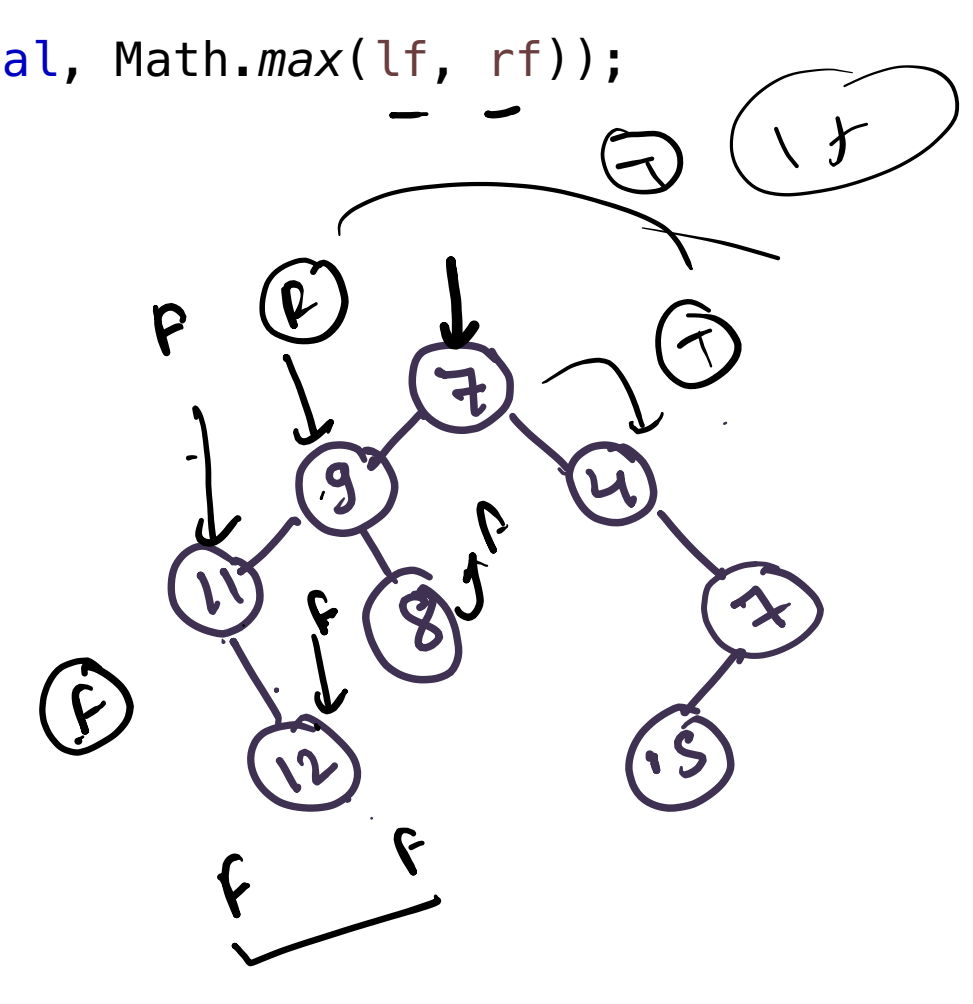
20 ←10→ 30
40 ←20→ .
. ←40→ .
. ←30→ 70
. ←70→ .


15
12
7
15
12
9
4
11
8
8
-∞
7
15
12
15
-∞
-∞
-∞
-∞

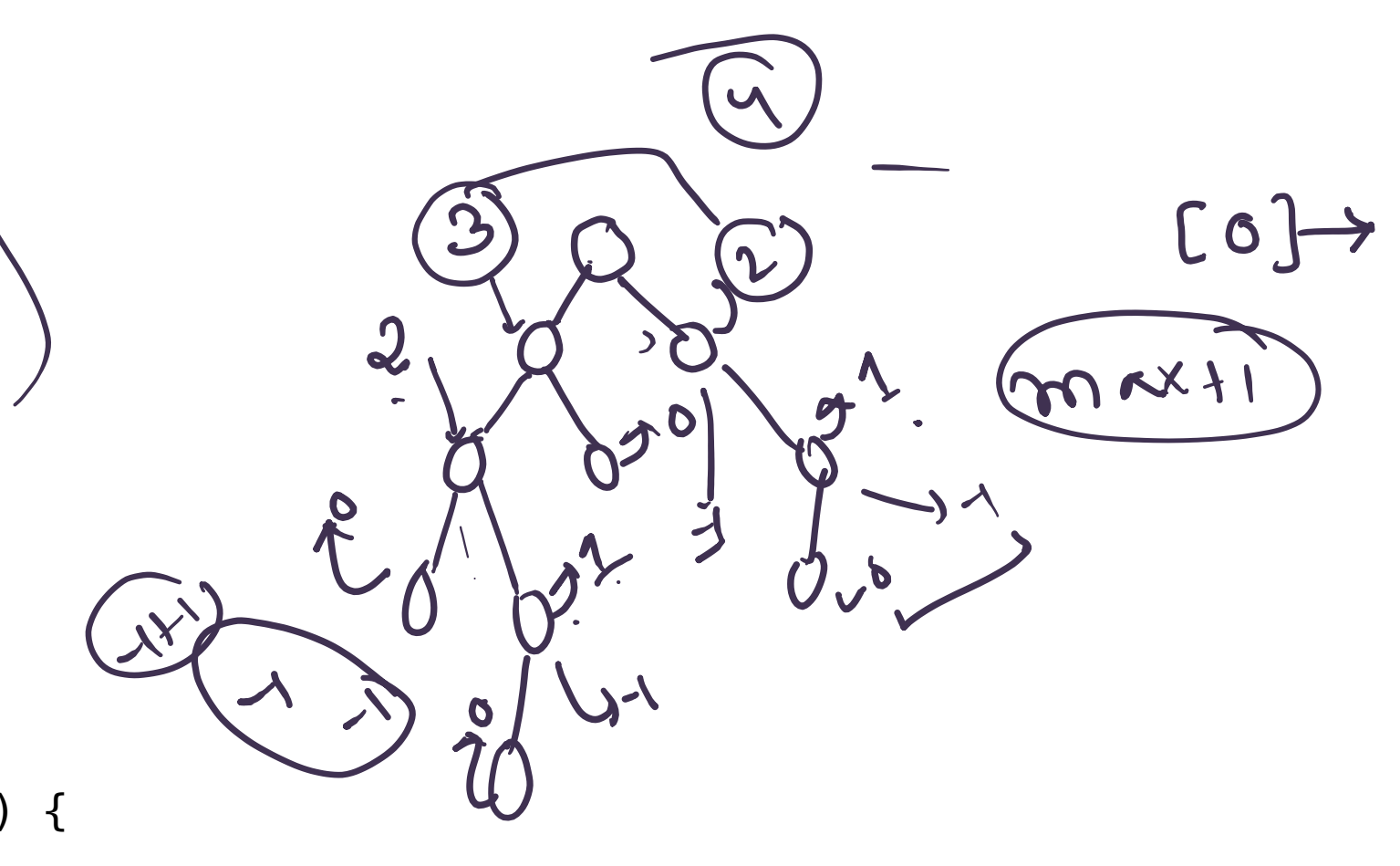
```
private int max(Node nn) {
    if(nn==null) {
        return Integer.MIN_VALUE;
    }
    int lf=max(nn.left);
    int rf=max(nn.right);
    return Math.max(root.val, Math.max(lf, rf));
}
```

```
private boolean find(Node nn, int val) {
    if(nn==null) {
        return false;
    }
    if(nn.val==val) {
        return true;
    }
    boolean l=find(nn.left, val);
    boolean r=find(nn.right, val);
    return l||r;

}
```


T
1 F
P
F
T
7
9
4
11
8
7
F
12
15
F
F

maximum dis b/w root to leaf Node


4
3
2
2
(-1-1)
-1 -1
1
0
1
0
1
-1
0
1
-1
0
-1

[0] → 1 ✓
0 ✓

max+1

```
private int ht(Node node) {
    if(node==null) {
        return -1;
    }
    int lh=ht(node.left);
    int rh=ht(node.right);
    return Math.max(lh, rh)+1;
}
```

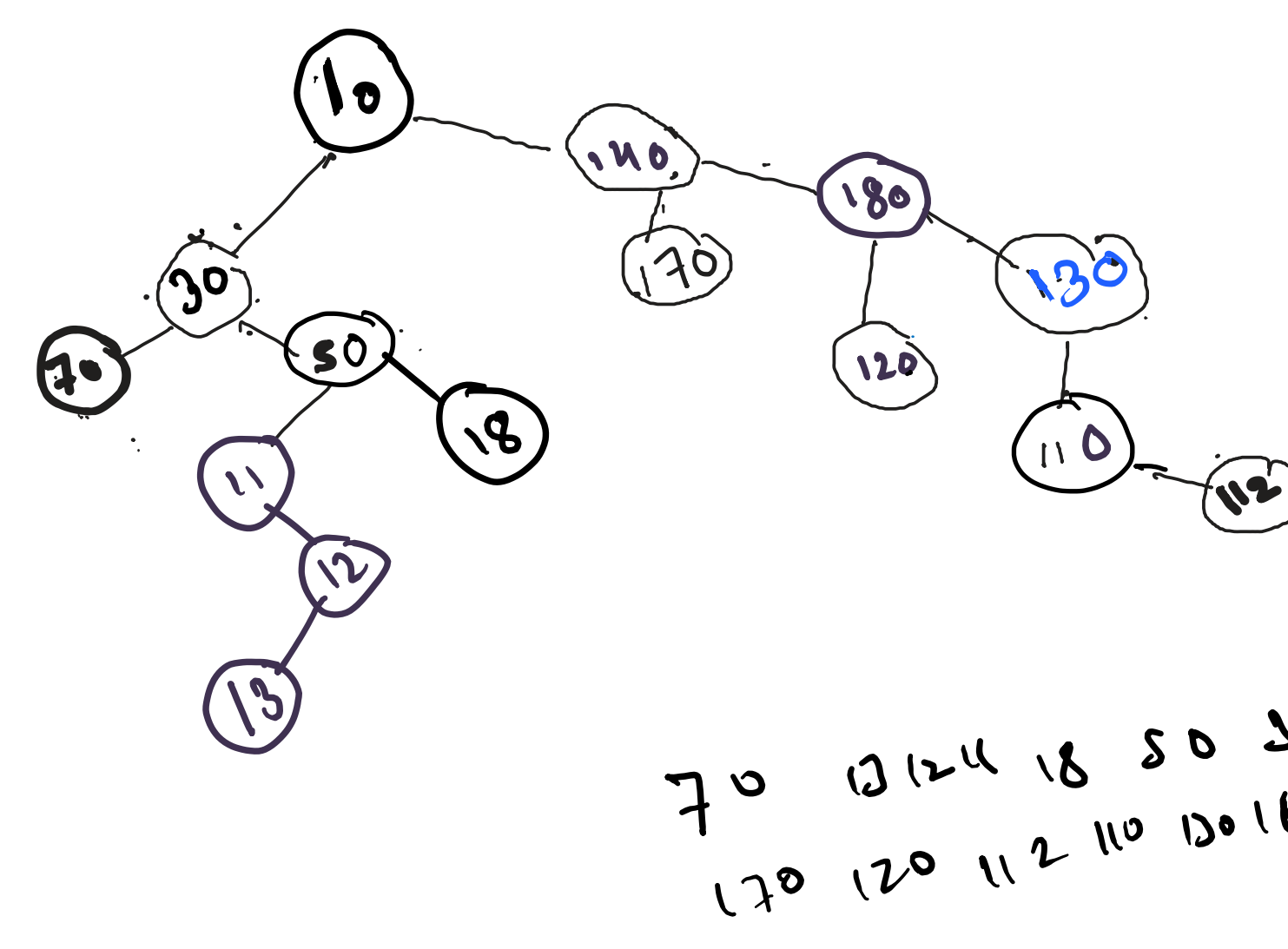
(i) pre-order → Root, Left, Right
(ii) In-order → Left Root, Right
(iii) Post-order → Left Right, Root
(iv) Level-order

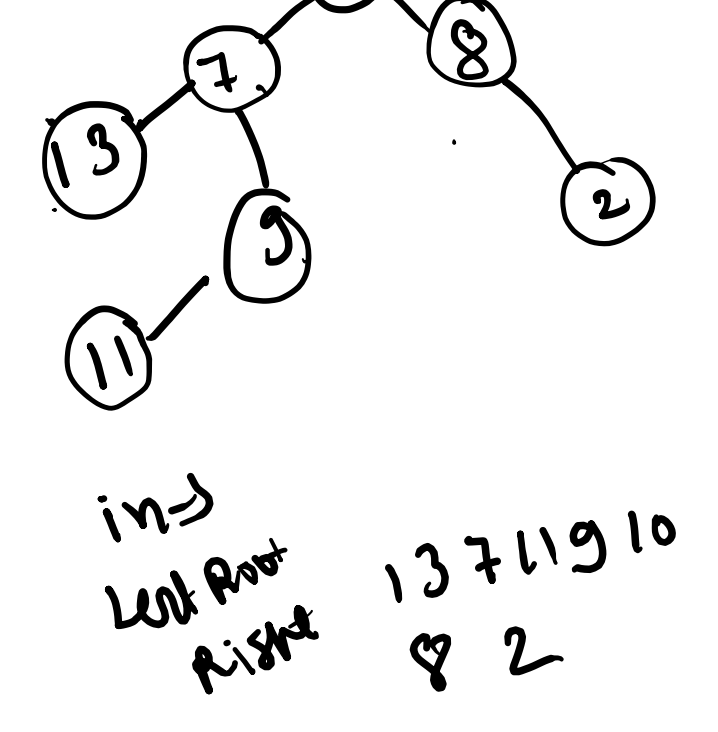
10 30 70 50
11 12 13 18 140 170
180 120, 130 110 112

70 30 11 13 12 50 18
10 170 140 120 180
110 112 130

Pre ← [10, 7, 13, 9, 11, 8, 2]
Root, Left
Right

Left Right Root
13 11 9 7 2 8 10

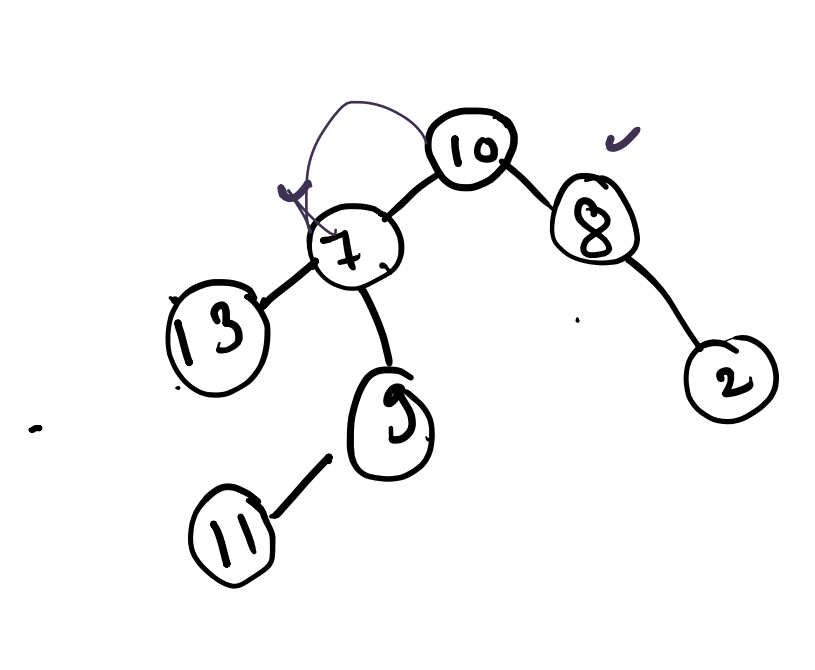

10
30
70
50
18
11
12
13
140
170
180
120
130
110
112


10
7
8
13
9
2
11

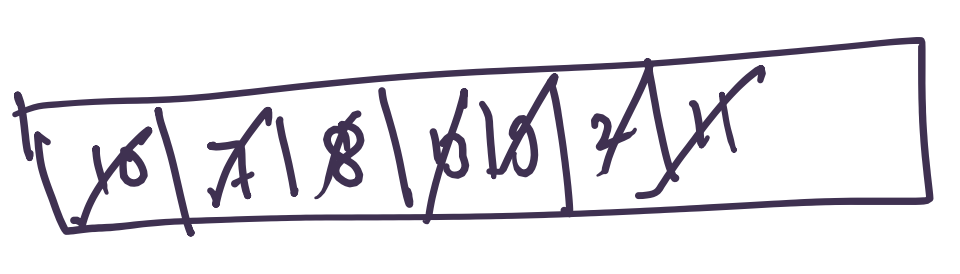
in →
Left Root
Right
13 7 11 9 10
8 2

70 13 12 11 18 50 30
170 120 112 110 130 180 140 10

Add last → O(1)
Rem F → O(1)


10
7
8
13
9
2
11

10 7 8 13 9 2 11


~~10~~
~~7~~
~~8~~
~~13~~
~~9~~
~~2~~
~~11~~

10 7 8 13 9 2